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RESEARCH TOPIC SELECTION SAMPLE

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Topic Selection Sample (Science) — Low-Carbon Process & Systems

Brand: ResearchEdit4U • Date: October 17, 2025

Disclaimer: Representative sample only. Topics use published literature for context. We cite sources accurately with DOIs. Replace/expand with client-specific literature during engagement.

Client Snapshot & Journey

Stage when contacting us	Idea stage (unclear scope), deadline in 3 weeks for supervisor check-in
Information provided	Field: Decarbonization/Process systems; Interests: carbon capture, hydrogen; Constraints: modelling only (no lab), focus on India/EU grids; Tools: Python available.
Researchedit4u solution	1) Elicited constraints & outcomes • 2) Proposed 3 refined topics • 3) Provided feasibility & reporting-fit • 4) Delivered supervisor-ready one-pager + appendices • 5) Created reading list with DOIs.
Client received	One-page Topic Pack, 2 appendices (rationales; concept maps), APA-7 references with DOIs, and a next-steps plan aligned to a 6-week sprint.

Refined Topic Options (2–3)

Topic 1: Hybrid Direct Air Capture (DAC) with Thermal Energy Storage (TES) for Grid-Interactive Operation

Problem/Gap: DAC is energy-intensive; few studies quantify how coupling with TES can reduce cost and emissions under renewable intermittency.

Why now: Grid variability + growing CDR demand require dispatchable negative emissions; decision-makers need co-optimised DAC+TES evidence.

Research Questions / Hypotheses:

• RQ1: How does TES sizing (sensible vs. latent) affect DAC cost per tCO_2 and net GHG under 24–72 h variability?

• RQ2/H1: Co-optimising fan speed and TES charge–discharge reduces specific energy consumption by ≥10%.

• RQ3/H2: In regions with ≥70% RES, cradle-to-gate LCA is net-negative (≥0.8 tCO_2 removed per t captured).

Topic 2: Electrified Steam Methane Reforming (eSMR) + CO_2 Capture for Blue Hydrogen on High-RES Grids

Problem/Gap: eSMR promises lower process emissions, but TEA+LCA vs. SMR+CCS under high-renewable grids is underexplored.

Why now: Emerging emissions thresholds for hydrogen require lifecycle clarity as grids decarbonise.

Research Questions / Hypotheses:

• RQ1: At which grid-carbon intensities does eSMR+CC capture outperform SMR+CCS on levelised cost and gCO_2e/MJ?

• RQ2/H1: Demand-response operation of eSMR yields ≥5% OPEX savings without breaching production constraints.

• RQ3: Sensitivity of results to capture rate (90–98%) and electricity price volatility.

Topic 3: Systematic Review & Meta-analysis — Calcium Looping in Cement: Performance, Degradation, and O&M Costs

Problem/Gap: Fragmented pilot data on sorbent deactivation and O&M costs impede bankable retrofits.

Why now: Cement decarbonisation requires synthesis of pilot learnings into generalisable parameters.

Research Questions / Hypotheses:

• RQ1: What are pooled capture efficiencies and degradation rates across pilot scales?

• RQ2/H1: Optimised purge strategies (≥15%) improve lifetime economics vs. baseline make-up rates.

• RQ3: What reporting gaps exist relative to PRISMA/EQUATOR for process trials? (PRISMA protocol to be pre-registered.)

Scope Box

Dimension	In Scope	Out of Scope
Population/System	Industrial energy systems (DAC; H_2 plant; cement kiln line)	Household devices
Setting	India/EU/US grids (scenario-based)	Off-world/lab-only w/o scale pathway
Time Window	2020–2035 modelling; 2015–2025 evidence for review	Pre-2000 tech w/o retrofit relevance
Variables	Energy, cost, capture rate, LCA intensity, reliability	Unrelated UX/marketing metrics

Feasibility Checks

✓ Data access: peer-reviewed datasets & models; vendor datasheets under NDA if needed.

✓ Ethics: minimal risk (modelling/review only); for surveys/interviews add STROBE/CONSORT alignment as relevant.

✓ Tools: Python/Matlab/ASPEN; LCA (Brightway/openLCA).

✓ Timeline realism: Topic pack 3–5 days; proposal skeleton in 7–10; full study 12–16 weeks.

Starter References (APA 7 with DOIs)

• Keith, D. W., Holmes, G., St. Angelo, D., & Heidel, K. (2018). A process for capturing CO_2 from the atmosphere. Joule, 2(8), 1573–1594. https://doi.org/10.1016/j.joule.2018.05.006

• House, K. Z., Baclig, A. C., Ranjan, M., van Nierop, E. A., Wilcox, J., & Herzog, H. J. (2011). Economic and energetic analysis of capturing CO_2 from ambient air. Proceedings of the National Academy of Sciences, 108(51), 20428–20433. https://doi.org/10.1073/pnas.1012253108

• Bui, M., Adjiman, C. S., Bardow, A., et al. (2018). Carbon capture and storage (CCS): The way forward. Energy & Environmental Science, 11, 1062–1176. https://doi.org/10.1039/C7EE02342A

• Mac Dowell, N., Fennell, P. S., Shah, N., & Maitland, G. C. (2017). The role of CO_2 capture and storage in the UK. Science, 356(6339), 1060–1064. https://doi.org/10.1126/science.aad0732

• Blamey, J., Anthony, E. J., Wang, J., & Fennell, P. S. (2010). The calcium looping cycle for large-scale CO_2 capture. Chemical Engineering Research and Design, 88(11), 1509–1517. https://doi.org/10.1016/j.cherd.2010.05.017

• Page, M. J., McKenzie, J. E., Bossuyt, P. M., et al. (2021). The PRISMA 2020 statement. BMJ, 372, n71. https://doi.org/10.1136/bmj.n71

• von Elm, E., Altman, D. G., Egger, M., et al. (2007). The STROBE statement. PLoS Medicine, 4(10), e296. https://doi.org/10.1371/journal.pmed.0040296

• Schulz, K. F., Altman, D. G., & Moher, D. (2010). CONSORT 2010 statement. BMJ, 340, c332. https://doi.org/10.1136/bmj.c332

PRISMA intent: If Topic 3 is chosen, we will pre-register a protocol and follow PRISMA for screening and reporting.

Appendix A — Topic Rationales

Topic 1

Hybrid DAC+TES directly tackles energy intensity and grid mismatch by shifting thermal/electric loads into low-carbon hours. This project co-optimises TES size and operating setpoints, reports cradle-to-gate LCA under realistic grid profiles, and outputs decision-grade curves ($/t$CO_2$ vs. %RES; net GHG/t). Feasible with public models and datasets; no human/animal subjects. Impact: actionable guidance for CDR developers and policymakers.

Topic 2

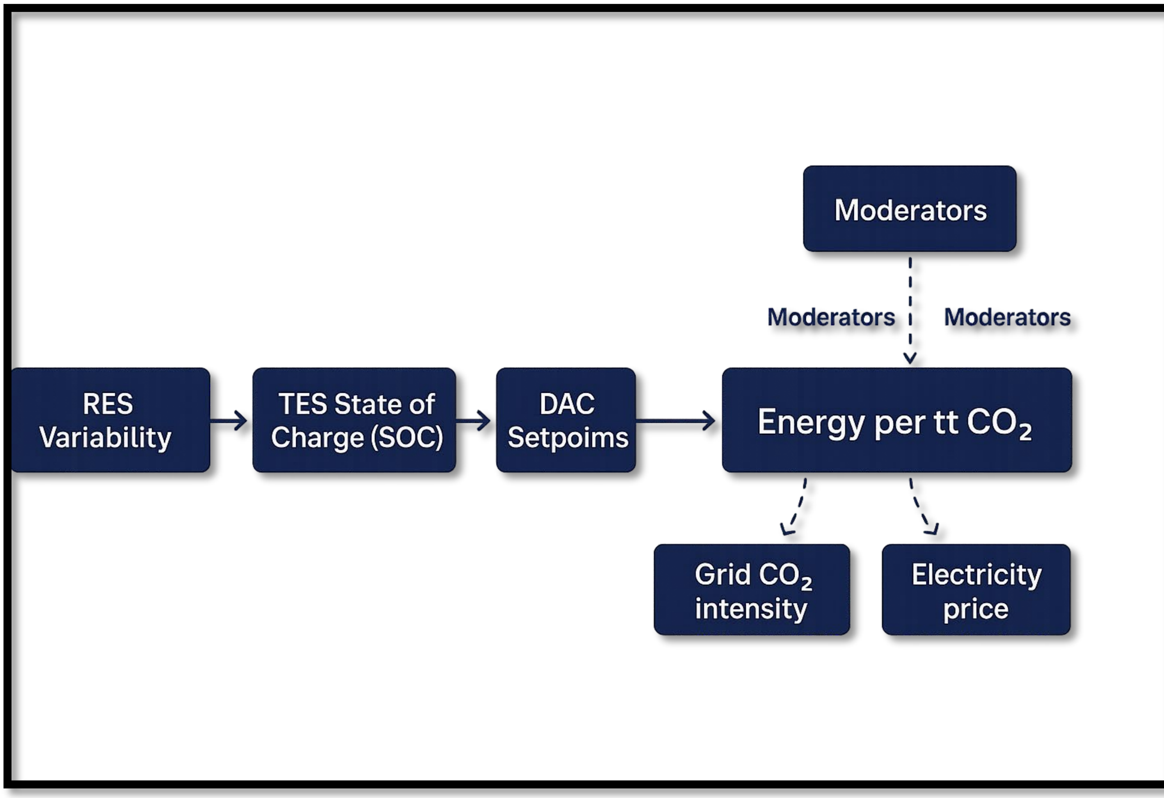
Electrifying process heat in SMR while capturing CO_2 could decouple emissions from combustion and leverage cleaner grids. We quantify crossover points for eSMR+CC vs. SMR+CCS on cost and gCO_2e/MJ, including demand-response operations. Feasible via mature SMR models and published capture performance. Impact: sequencing guidance for blue vs. green H_2 in 2030–2040 transitions.

Topic 3

Pilots of calcium looping report divergent sorbent degradation and O&M data. A PRISMA-based synthesis with meta-analysis will pool capture efficiency and make-up rates, standardise reporting, and map evidence gaps. Feasible (published data only). Impact: bankable parameters for TEA and standards bodies.

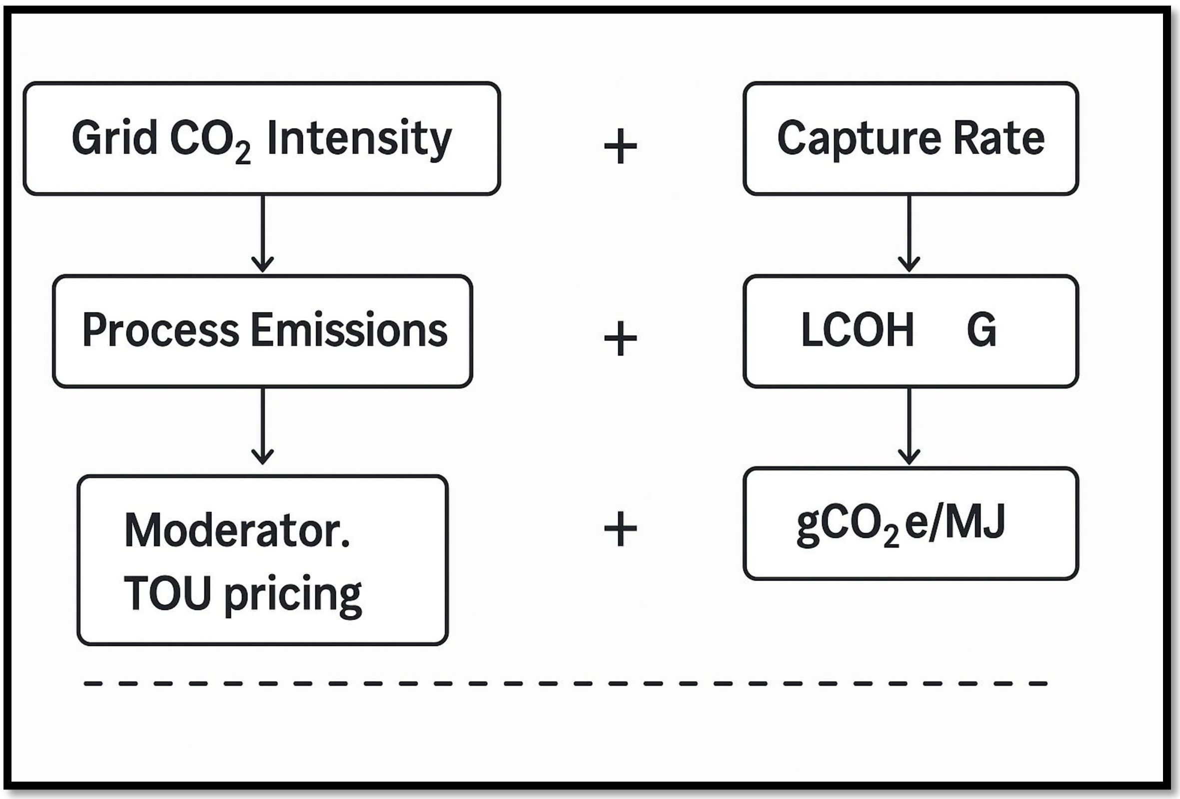
Appendix B — Concept Maps & Reporting Fit

Topic 1 Concept Map



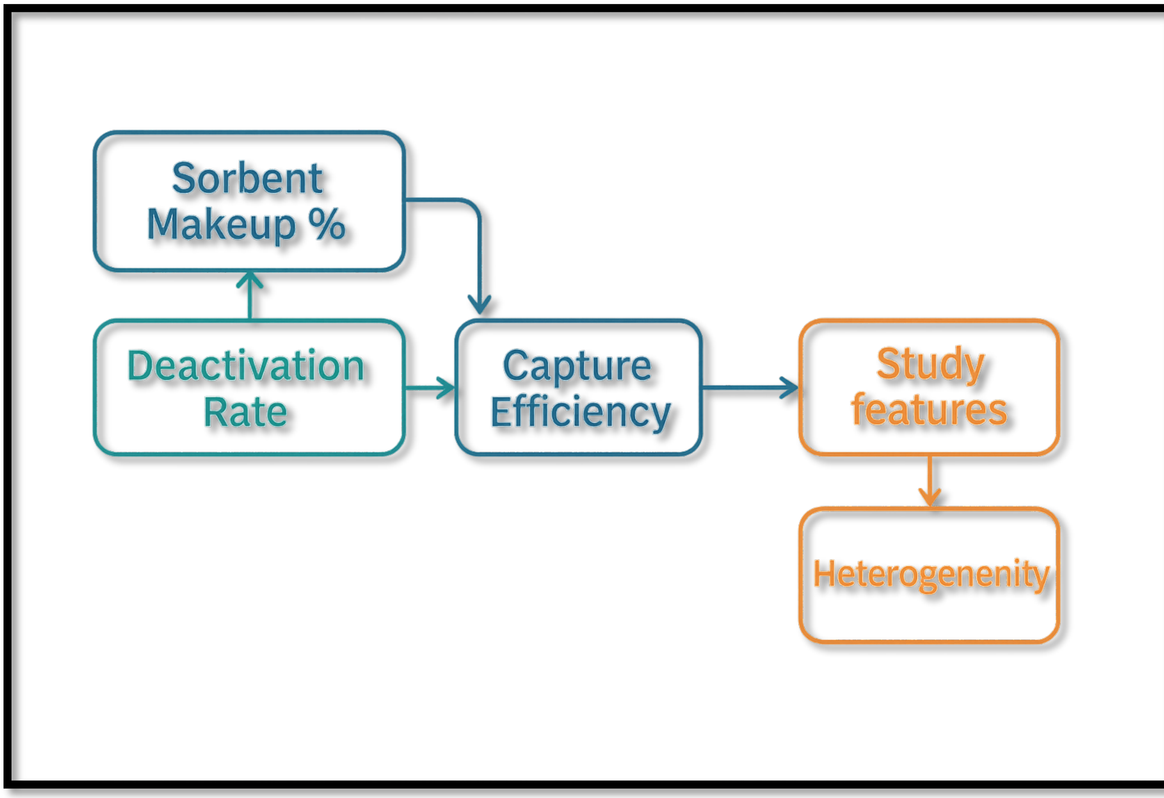
[RES Variability] → [TES SOC] → [DAC Setpoints] → [Energy/tCO_2] → [Cost & Net GHG]; Moderators: Grid CO_2 intensity, price.

Topic 2 Concept Map



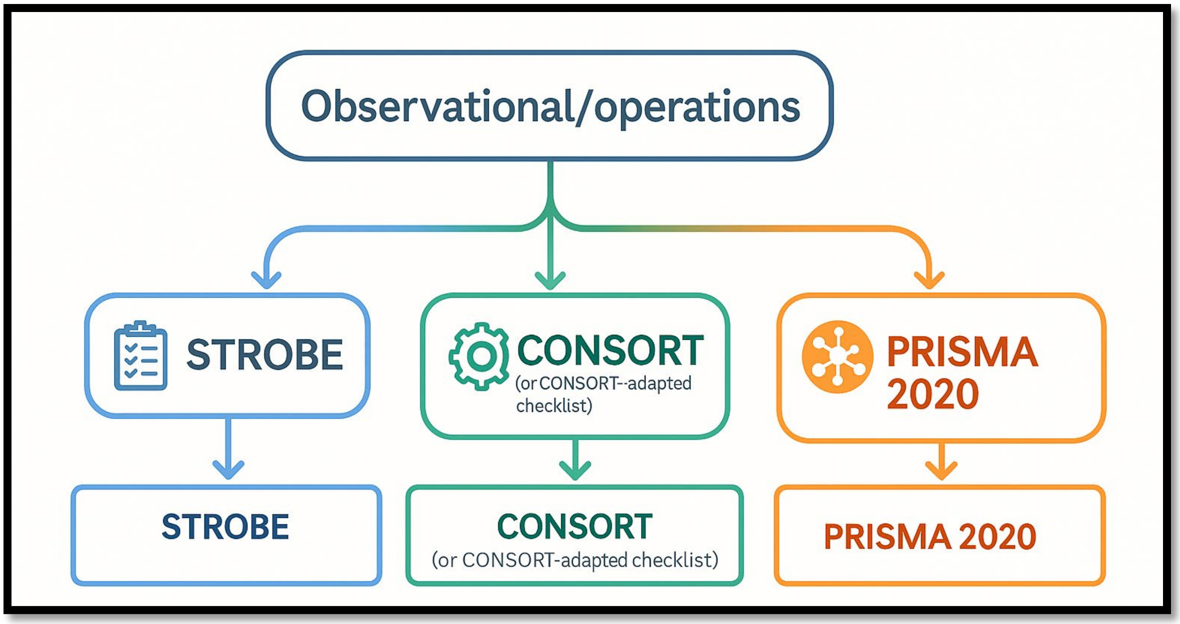
[Grid CO_2 Intensity] + [Electrifier η] → [Process Emissions]; [Capture Rate] & [Penalty] → [LCOH] & [gCO_2e/MJ]; Moderator: TOU pricing.

Topic 3 Concept Map



[Sorbent Makeup %] ↔ [Deactivation Rate] → [Capture Efficiency] → [O&M Cost/tCO_2];
Study features → Heterogeneity → Meta-moderators.

Reporting-fit (EQUATOR family)



Observational/operations → STROBE; process trials → CONSORT (or CONSORT-adapted checklist); systematic review/meta-analysis → PRISMA 2020.

THANK YOU

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